Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

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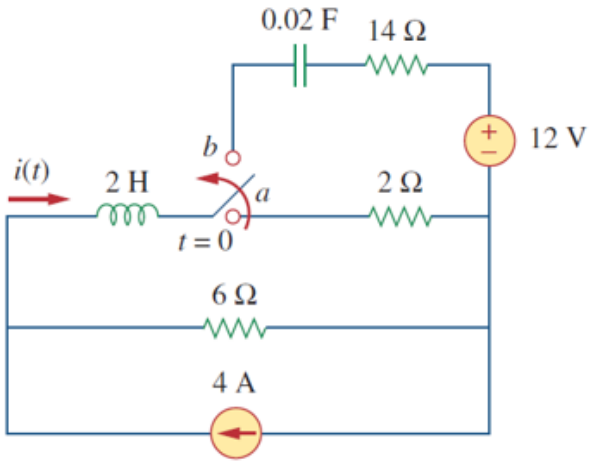
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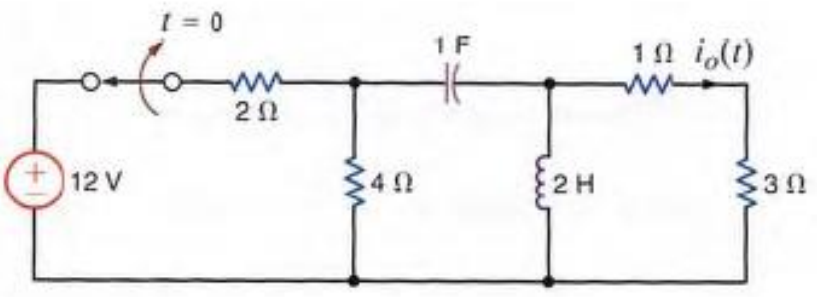
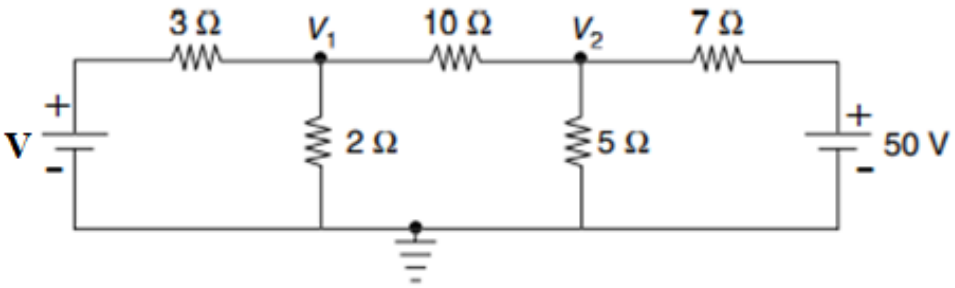
THIRD SEMESTER BTECH. (E & C) DEGREE PROCTORED ONLINE EXAMINATION
JANUARY 2022
SUBJECT: NETWORK ANLYSIS (ECE - 2154)

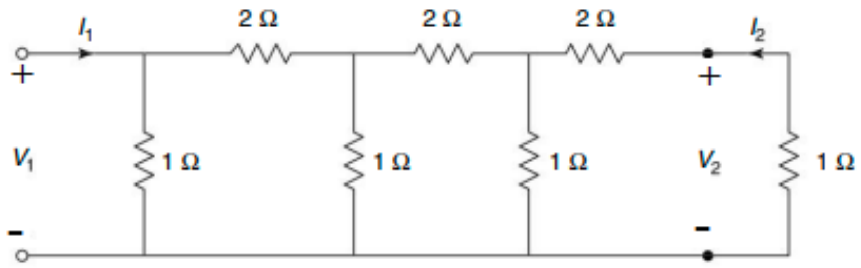
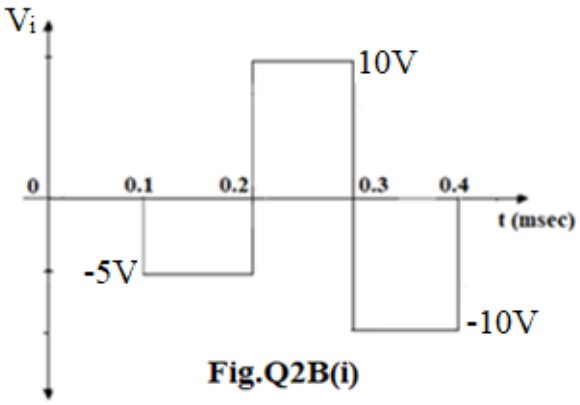
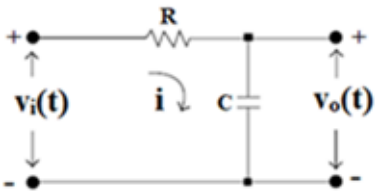
TIME: 75 min (9.20 AM to 10.35 AM) **MAX. MARKS: 20**

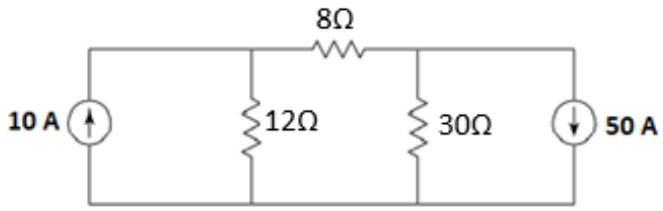
Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.
- Do not use **Laplace Transforms** unless specified.
- Answer script must be a single pdf, good visibility of all texts and numbers.
- Name the file as: RegistrationNumber_SubjectCode_dd_JAN2022
- Upload correct pdf properly named well before 10.45 AM

Q. No.	Questions	Marks
1A.	In the circuit shown Fig.Q1A, the switch is moved from position a to position b at t=0. Find i(t) for $t > 0$.  **Fig.Q1A**	4

1B.	In the circuit shown in Fig.Q1B, the switch is opened at t=0. Find $i_o(t) for\ t > 0$ using Laplace Transform.  **Fig.Q1B**	3
1C.	Determine the voltage V in the network shown in Fig.Q1C which makes the current in the 10 ohms resistor zero.  **Fig.Q1C**	3

2A	Determine the Network Functions *(i)* $\frac{V_2}{V_1}$ $(ii)\frac{V_2}{I_1}(iii)$ $\frac{I_2}{V_1}$ and $(iv)\frac{I_2}{I_1}$ for the network shown in Fig.Q2A.  **Fig.Q2A**	4
2B.	The waveform shown in Fig.Q2B(i) is applied on to the circuit shown in Fig.Q2B(ii). Calculate and sketch the output waveform. Consider time constant = 0.1msec.  **Fig.Q2B(i)**  **Fig.Q2B(ii)**	3

2C	For the circuit shown in Fig.Q2C, determine the current through the 8Ω resistor using Superposition theorem.  **Fig.Q2C**	3